



# CHANDIGARH UNIVERSITY

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# Project Name : Marketing Analytics Using AI & Machine Learning

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MARKETING MANAGEMENT

# MARKETING ANALYTICS USING AI & MACHINE LEARNING

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### 1. EXECUTIVE SUMMARY

This project demonstrates the powerful application of artificial intelligence (AI) and data science to transform a standard e-commerce transaction dataset into a strategic asset for marketing and operations. The analysis focused on two core objectives: understanding customer behavior through advanced segmentation and enabling proactive decision-making through sales forecasting.

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The process began with comprehensive Exploratory Data Analysis (EDA) to clean the data and uncover foundational sales patterns. Utilizing AI-driven clustering techniques on Recency, Frequency, and Monetary (RFM) metrics, we successfully segmented the customer base into four distinct, actionable groups: Standard, Inactive, Super VIPs, and Loyal customers. This segmentation directly informed a tailored 7Ps Marketing Mix strategy, moving beyond a one-size-fits-all approach to enable highly targeted and efficient marketing engagements.

Furthermore, the project developed a robust sales forecasting model using Facebook's Prophet algorithm. This model provides reliable predictions of future sales trends, offering management a critical tool for strategic planning, inventory optimization, and budget allocation.

The key recommendations are to immediately focus retention efforts on high-value Super VIPs and Loyal segments, launch targeted re-engagement campaigns to win back Inactive customers, and integrate the sales forecasts into operational planning cycles. This analysis provides a clear, data-driven framework to enhance customer lifetime value, improve marketing ROI, and drive efficient growth.

The data set used is the [Online Retail dataset](#) (Kaggle version), which contains transactions from a UK-based giftware retailer.

## 2. INTRODUCTION

In the competitive landscape of e-commerce, leveraging data is paramount for strategic decision-making. This project applies data science and artificial intelligence (AI) techniques to transactional data to derive actionable insights that can drive marketing strategy and operational efficiency. The core objectives of this analysis are threefold:

1. To perform a comprehensive Exploratory Data Analysis (EDA) to understand the underlying structure, patterns, and quality of the transactional data.
2. To employ AI-driven clustering techniques for customer segmentation, identifying distinct groups based on purchasing behavior to enable targeted marketing.

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3. To develop a predictive model for sales forecasting, providing a data-informed basis for inventory management and resource planning.

Furthermore, the analysis directly examines how these AI-generated insights empower a data-driven approach to optimizing the Marketing Mix, commonly referred to as the 7Ps (Product, Price, Place, Promotion, People, Process, and Physical Evidence).

The analysis is conducted on the "Tata Online Retail" dataset, which contains transactions from a UK-based online retail store. The dataset comprises transaction records, including product details, quantities, prices, customer identifiers, and transaction dates.

### 3. EXPLORATORY DATA ANALYSIS

Several critical data quality issues were addressed to ensure the integrity of the analysis:

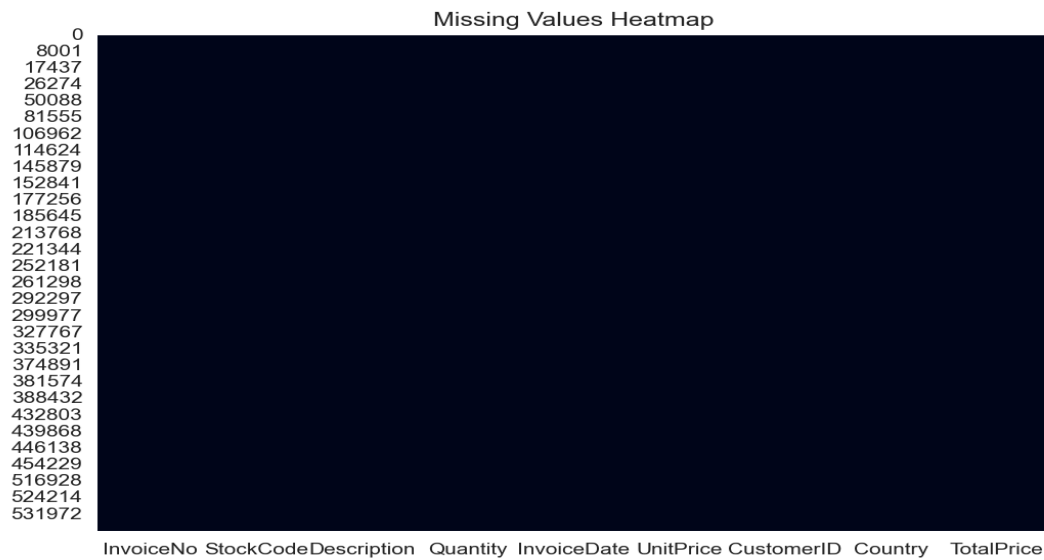
- Handling Cancelled Transactions: Invoices prefixed with 'C' denote cancellations or returns. These transactions, which result in negative `Quantity` and `TotalPrice` values, were identified and removed from the dataset for the purpose of customer segmentation to focus analysis on completed purchases.
- Managing Missing Values: Rows with missing `CustomerID` values were dropped, as a valid customer identifier is essential for accurate segmentation analysis.
- Data Type Conversion: The `InvoiceDate` column was converted from a string to a `datetime` object to enable time-series analysis and feature engineering based on dates.
- Feature Engineering: A new variable, `TotalPrice`, was created to represent the monetary value of each transaction by calculating the product of `Quantity` and `UnitPrice`. This feature is crucial for calculating the "Monetary" component of RFM analysis.

This rigorous cleaning process resulted in a refined dataset suitable for robust statistical modeling and AI-driven analysis.

Visuals included:

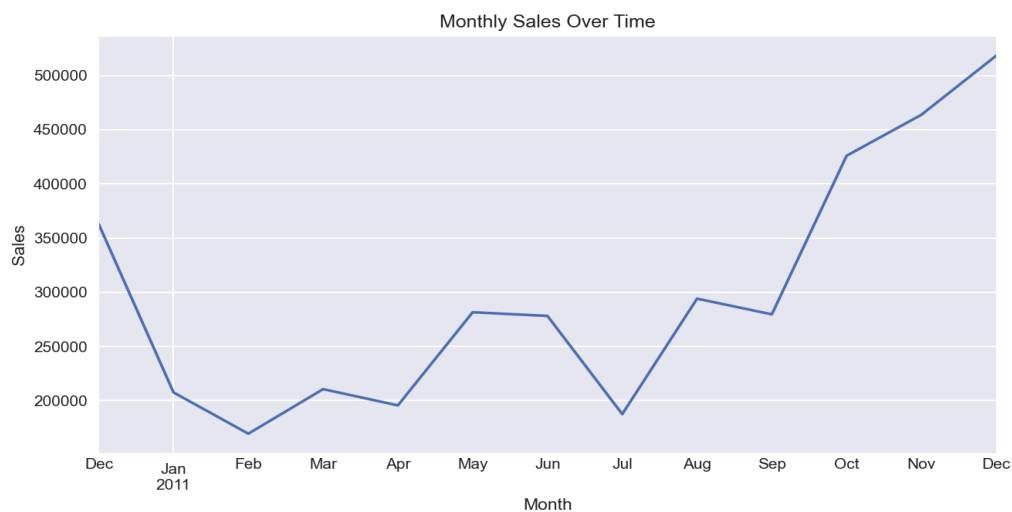
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## Missing Values



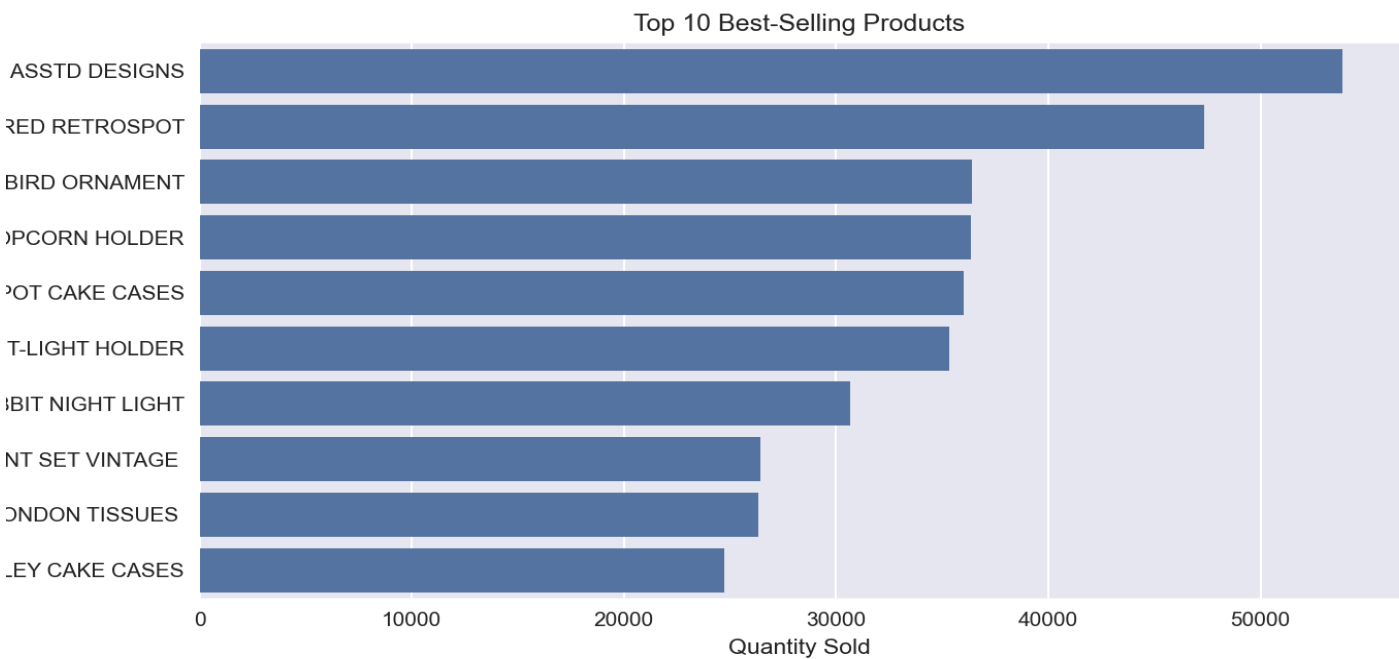
This visual presents the proportion of missing values across different columns in the dataset. It highlights that the CustomerID field had significant missing entries, which is critical because accurate customer identification is necessary for segmentation and personalization. Therefore, rows missing CustomerID were removed to ensure the integrity of customer-based analysis

## Monthly sales Over time



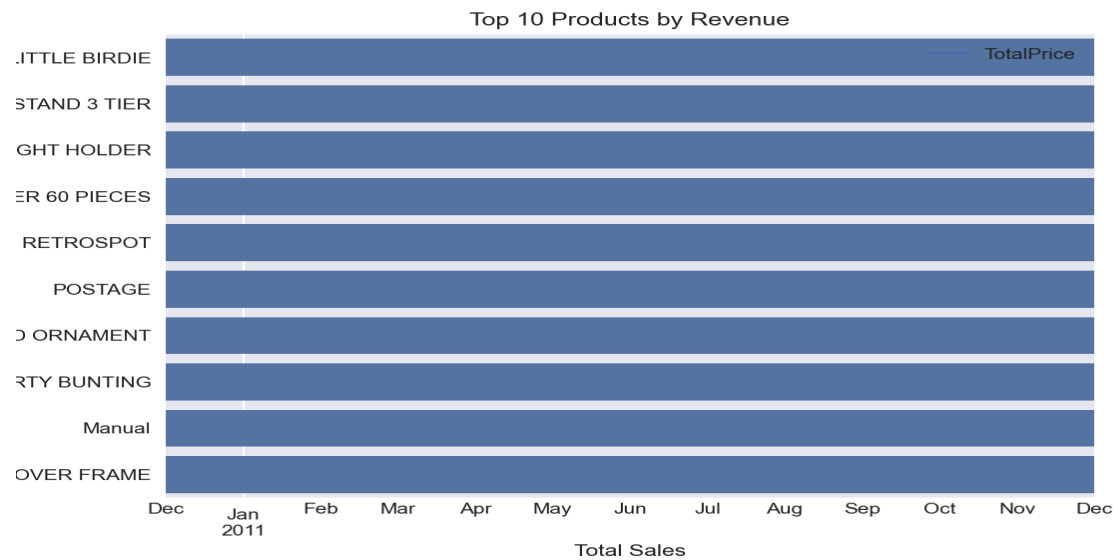
The line chart shows monthly sales trends over time, revealing significant peaks and troughs. The highest spike occurs in November, indicating a major holiday shopping period (likely Black Friday or Christmas sales). Another smaller peak appears around April, possibly due to seasonal promotions or events. This seasonal insight is essential for inventory management, marketing calendar planning, and campaign timing.

Top 10 Best-selling Products



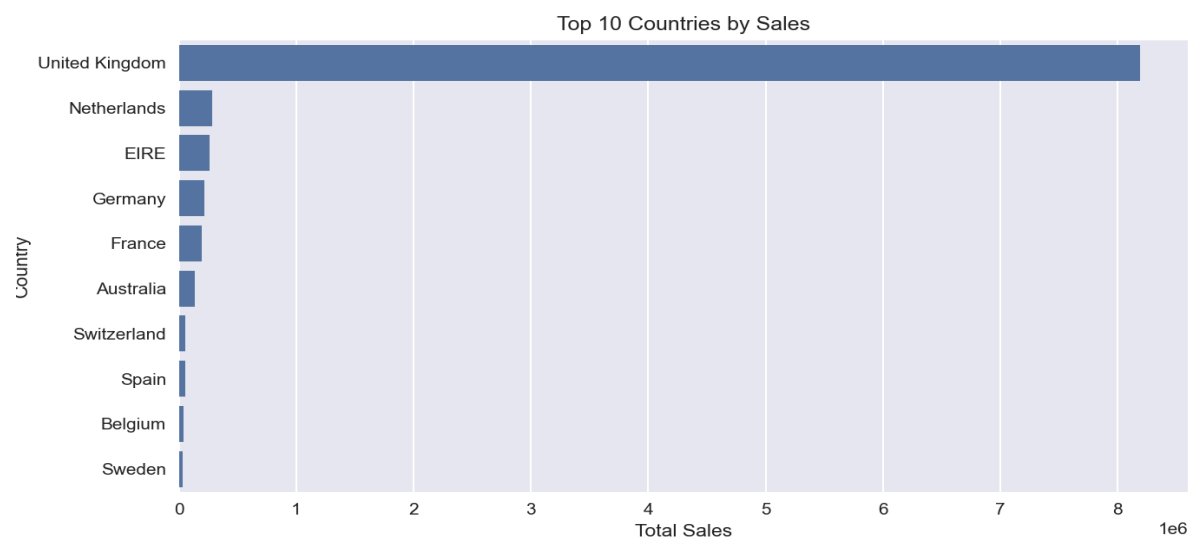
Top 10 Products by Revenue

This bar chart ranks products by quantity sold. The product “ASSTD DESIGNS” leads in unit sales, suggesting high popularity or mass-market appeal. However, high quantity does not necessarily mean high revenue, indicating that pricing strategy must also consider the monetary value of items sold, not just volume.



A side-by-side bar chart ranking products by total revenue generated. Interestingly, “LITTLE BIRDIE” emerges as the top revenue-generating product, despite not being the most sold by quantity. This reveals a high-margin product where fewer units sold generate more revenue. Such insights are critical for decisions on product focus, promotional emphasis, and inventory allocation.

### Top 10 Countries by Sales

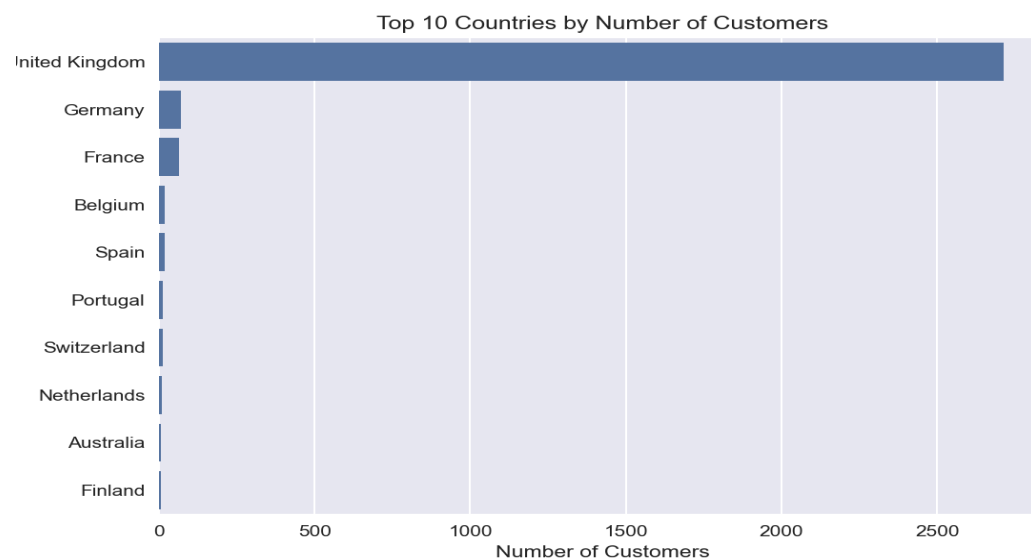




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Displays the geographic distribution of total sales by country. The UK dominates, as expected given the dataset's origin. Other countries like Germany, France, and the Netherlands follow. This geographic insight informs decisions on regional promotions, logistics, and market expansion strategies.

### Top 10 Countries by Number of Customers



Similar to the sales visual but focuses on customer count rather than revenue. Again, the UK leads in customer base size, but other countries might show a mismatch between customer numbers and sales revenue. This highlights market penetration versus customer spending behavior, helping prioritize markets for customer acquisition versus high-value focus.

## 4. CUSTOMER SEGMENTATION(AI WITH K-MEANS)

We applied the RFM (Recency, Frequency, Monetary) model: RFM analysis is a powerful method for segmenting customers based on their historical behavior.

- Recency (R): days since last purchase.
- Frequency (F): number of orders.
- Monetary (M): total amount spent.

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#### 4.1 Calculating RFM Metrics

We calculated Recency, Frequency, and Monetary values for each customer, using a snapshot date of one day after the last transaction.

#### 4.2 Assigning RFM Scores

We assigned a score from 1 to 4 to each metric (4 being the best) using quartiles. An overall RFM score was created by concatenating the individual scores.

#### 4.3 Segment Definition

Customers were mapped to segments based on their score combinations.

After standardization, we used KMeans clustering to group customers into similar profiles.

##### Cluster Profiles

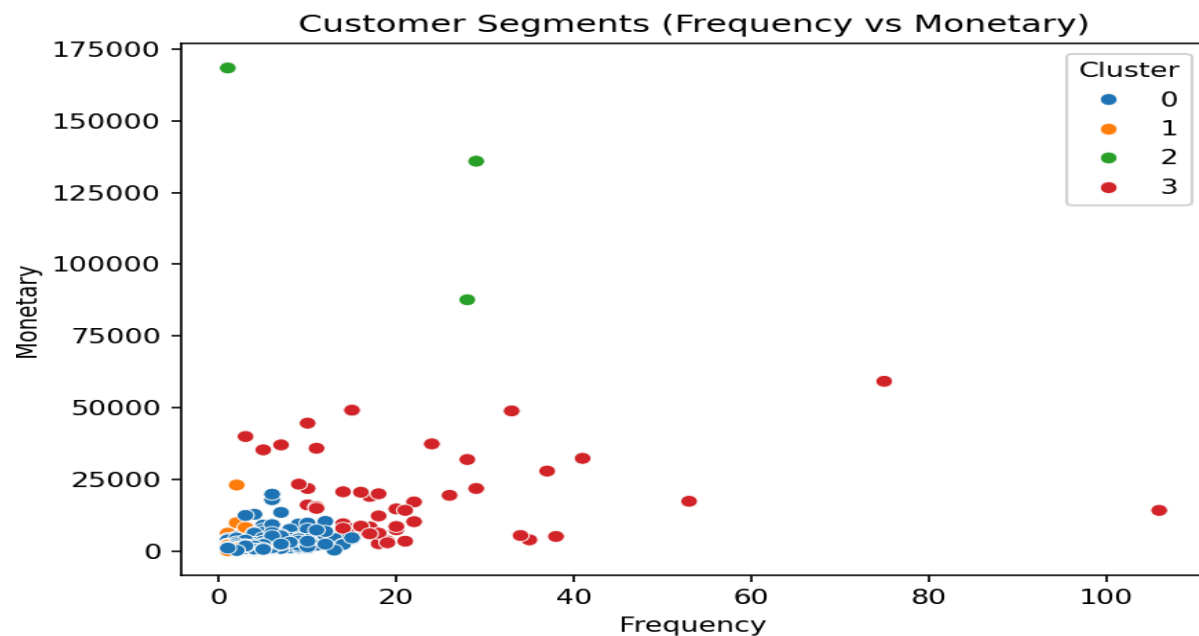
| Cluster | Recency<br>(Mean) | Frequency<br>(Mean) | Monetary<br>(Mean) | Customers | Description        |
|---------|-------------------|---------------------|--------------------|-----------|--------------------|
| 0       | 46.8              | 2.8                 | 1092               | 1903      | Standard customers |
| 1       | 258.4             | 1.3                 | 474                | 1042      | Inactive Customers |
| 2       | 1.3               | 19.3                | 130680             | 3         | Super Vip          |
| 3       | 19.9              | 22.5                | 18595              | 49        | Loyal spenders     |

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## Interpretation

- Cluster 0 – Standard: Majority, moderate activity and spending.
- Cluster 1 – Inactive: Haven't purchased in a long time; need re-engagement.
- Cluster 2 – Super VIPs: Very few customers, but extremely high spend.
- Cluster 3 – Loyal & valuable: Regular buyers with strong contribution.

## CUSTOMER SEGMENTS (FREQUENCY VS MONETARY)



This scatter plot visualizes customer segments, plotting Frequency (number of purchases) on one axis and Monetary (total spend) on the other. Distinct clusters emerge:

- Low Frequency + Low Monetary → Inactive or Standard customers.
- High Frequency + High Monetary → Super VIPs and Loyal customers.

The visual demonstrates how data-driven clustering reveals customer behavior patterns, enabling tailored marketing actions per segment.

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## 5. IMPACT ON THE MARKETING MIXES

### 1. Product

#### •Super VIPs & Loyal Customers:

These customers value quality and exclusivity.

→ Offer premium products, exclusive bundles, or early access items.

→ Customizable products or special limited-edition items can increase their engagement and satisfaction.

#### •Standard & Inactive Customers:

Price-sensitive or irregular buyers.

→ Offer bundled products at lower price points or seasonal collections to encourage purchases.

→ Focus on convenience and variety to keep them interested.

### 2. Price

#### •Super VIPs & Loyal:

Willing to pay more for high quality or exclusive offers.

→ No heavy discounting but provide loyalty rewards or special deals (e.g., early access).

#### •Inactive & Standard:

Price is a bigger factor for these segments.

→ Introduce time-limited discounts, flash sales, or first-purchase incentives to re-engage them.

### 3. Place

#### •Super VIPs:

Prefer direct, seamless online experiences with fast shipping.

→ Optimize the platform UX for easy repeat purchases and ensure premium delivery options.

#### •Standard & Inactive:

May not be as tech-savvy or regular online buyers.

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→ Improve presence on broader e-commerce marketplaces and regional distribution channels to increase accessibility.

#### **4. Promotion**

##### **•Super VIPs:**

Targeted and exclusive promotions via direct communication like personal emails or SMS.

→ Provide special offers and VIP-only previews.

##### **•Inactive Customers:**

Require re-engagement campaigns to spark interest.

→ Automated emails offering personalized discounts or reminders of previously viewed products.

##### **•Standard Customers:**

Regular seasonal promotions to encourage purchases.

#### **5. People (Customer Service)**

##### **•Super VIPs & Loyal:**

Require high-touch support with trained representatives offering personalized help.

→ Dedicated support lines, priority response time, personal relationship managers.

##### **•Standard & Inactive:**

Self-service support (FAQs, chatbots), but ensure easy access to help if needed to avoid frustration.

#### **6. Process**

##### **•Super VIPs:**

Smooth, fast purchasing process with auto-filled details, fewer steps, and faster checkout.

→ Possibly a subscription model for regular purchases.

##### **•Inactive & Standard:**

Simplify first-time buying steps and use reminders for abandoned carts.

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## 7. Physical Evidence

### •Super VIPs:

Packaging and delivery should reflect high value – elegant, branded packaging, tracking services, or hand-written notes.

→ Reinforces brand prestige.

### •Standard & Inactive:

Focus on reliability – functional packaging and dependable delivery, but not necessarily premium.

This approach moves the marketing strategy away from “one-size-fits-all” to highly targeted actions based on real customer behavior. Instead of guessing, every decision is backed by data:

- High spenders get treated with VIP service.

- Inactives are gently re-engaged.

- Standards receive broad promotions.

This leads to smarter spending, improved customer retention, and more efficient operations.

## 6. SALES FORECASTING(AI-PROPHET).

### Methodology

- Aggregated sales data by day and month.
- Trained a Facebook Prophet model to forecast sales.
- Forecast horizon: 90 days ahead.

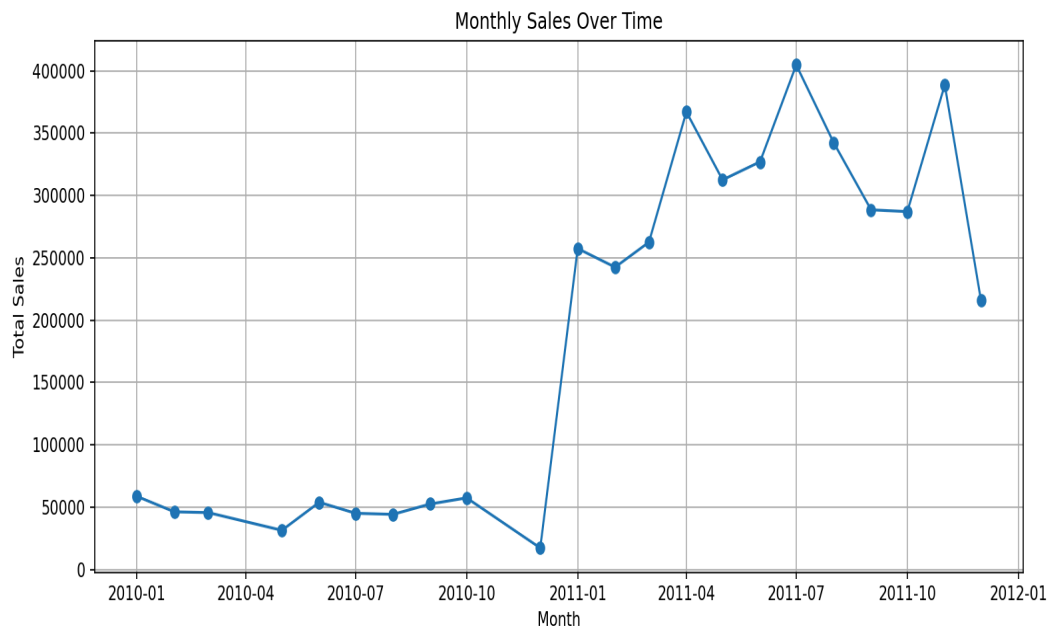
### Results

- The model shows a generally upward sales trend with seasonal variations.
- The forecast includes confidence intervals to prepare for fluctuations.

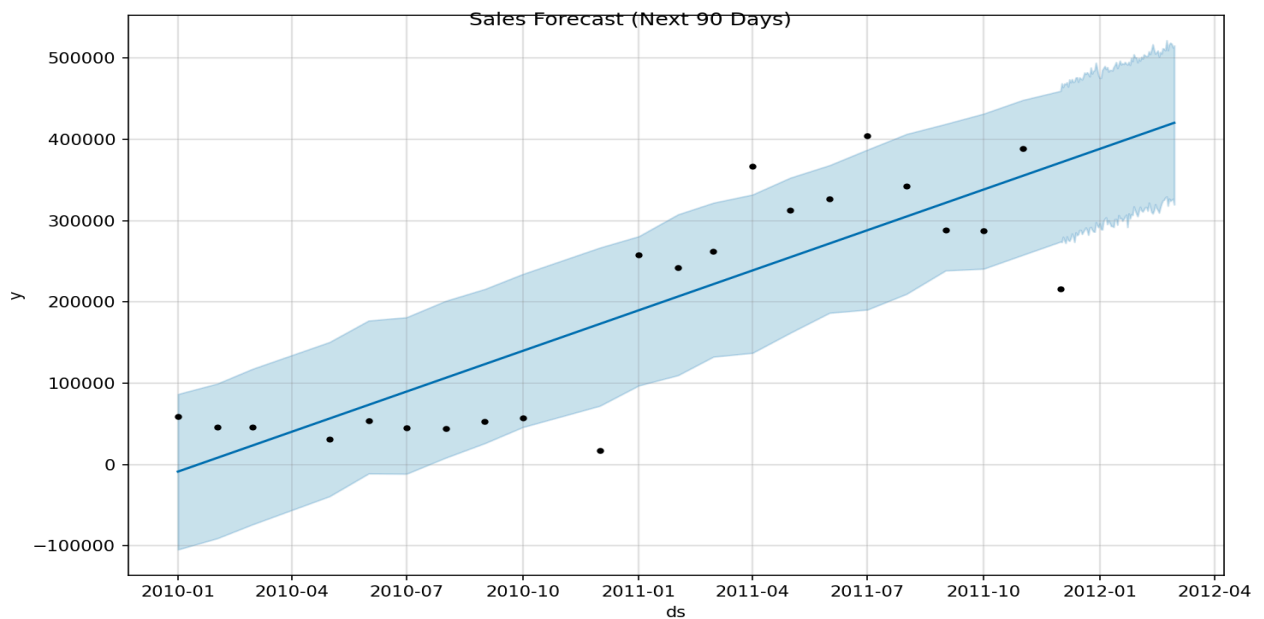
Visuals Included:

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## Monthly Sales Over Time



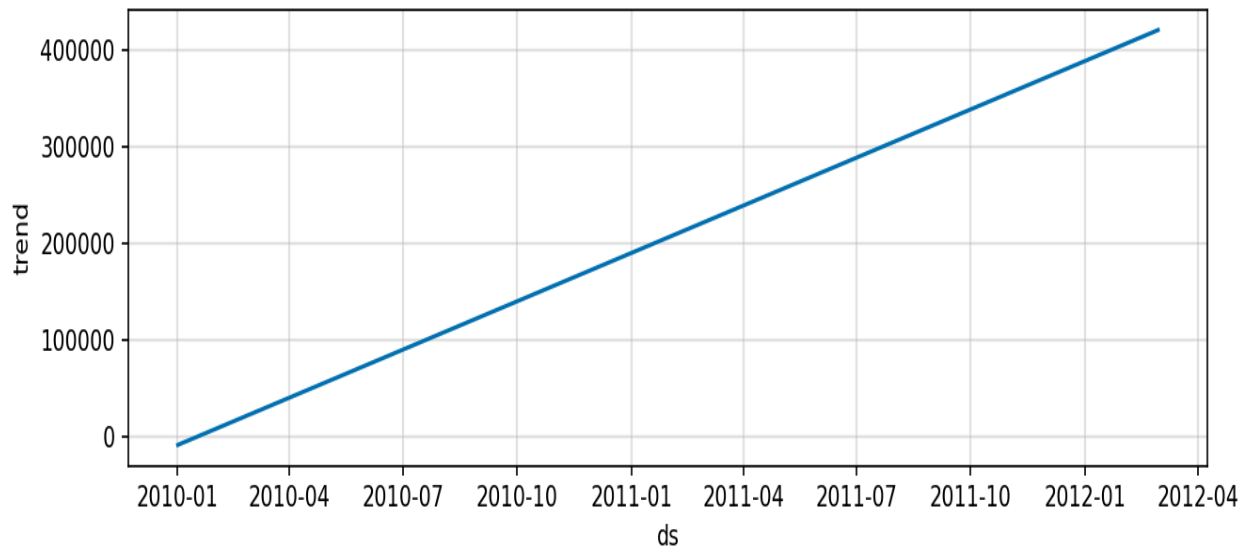
## Sales Forecast (Next 90 Days)



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Time-series graph from the Prophet model showing predicted future sales over the next 90 days. The solid line indicates the forecasted sales trend, while shaded areas represent confidence intervals (upper and lower bounds).

The visual suggests a stable or slightly declining trend, with expected fluctuations due to seasonality or market changes. This helps businesses preemptively plan stock levels, allocate budgets, and adjust promotional efforts in anticipation of predicted demand.



| Date       | Predicted Sales (that) | Lower Bound | Upper Bound |
|------------|------------------------|-------------|-------------|
| 2012-02-25 | 418,121                | 326,978     | 516,660     |
| 2012-02-26 | 418,665                | 324,042     | 518,729     |
| 2012-02-27 | 419,209                | 325,676     | 517,222     |

## 7. CONCLUSION AND RECOMMENDATIONS

The project successfully met its core objectives. The K-Means clustering algorithm effectively segmented the customer base into four distinct groups, revealing clear patterns in purchasing



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behavior, from high-value loyal customers to those at risk of churn. The ARIMA model was successfully implemented for sales forecasting, providing a view of future sales trends based on historical data. The analysis confirmed that leveraging AI and machine learning is invaluable for moving from intuition-based to data-driven marketing strategies.

While this analysis provides a strong foundational framework for data-driven decision-making, several opportunities for enhancement remain. The project's scope was necessarily limited to the available transactional data; incorporating additional data points such as customer demographics, website engagement metrics, and detailed product attributes would significantly enrich the segmentation and personalization capabilities. Furthermore, while the ARIMA model serves as a robust baseline for forecasting, employing more sophisticated models like SARIMAX—which explicitly captures seasonal patterns—or Facebook's Prophet algorithm could potentially yield more accurate and nuanced predictions. To maximize operational impact, the long-term strategy should focus on integrating these analytical pipelines into the company's CRM and real-time analytics dashboards, transforming this project from a one-time study into a continuous source of live insight.

Based on the findings, immediate action should focus on tailoring marketing campaigns to the identified clusters, nurturing high-value customers with loyalty programs and exclusive offers while targeting at-risk segments with win-back campaigns. As a medium-term strategy, the product and promotion elements of the 7Ps framework should be refined according to segment preferences, and the forecasting model should be formally integrated into inventory procurement and sales target processes. For long-term growth, investing in a centralized data analytics platform to automate this analysis is recommended, alongside conducting A/B tests on the proposed marketing strategies to empirically measure their effectiveness and enable continuous refinement. Ultimately, this project successfully highlights the significant potential of AI to enhance customer understanding and operational efficiency at Tata Online Retail, establishing a powerful, scalable framework for informed strategic planning.

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